

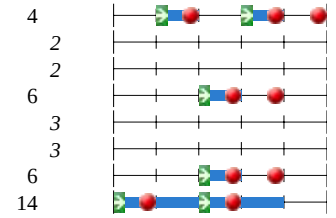
Program overview

08-Dec-2022 11:30

Year 2022/2023
Organization Mechanical, Maritime and Materials Engineering
Education Bachelor Mechanical Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Bachelor WB 2022							
1e jaar BSc WB 2022	1st Year BSc WB 2022						
Wiskunde blok							
WBMT1050	Mathematics 1	6					
WBMT1050 Toets 1	Analyse 1 - deeltentamen	3					
WBMT1050 Toets 2	Analyse 2 - deeltentamen	3					
WBMT1051	Mathematics 2	6					
WBMT1051 Toets 1	Lineaire Algebra 1 - deeltentamen	3					
WBMT1051 Toets 2	Lineaire Algebra 2 - deeltentamen	3					
Werktuigbouwkunde Theorie blok							
WB1135	Introductory Dynamics	6					
WB1530-14	Thermodynamics	6					
WB1630-20	Statics	6					
WB1631-15	Mechanics of Materials	6					
Werktuigbouwkunde Project blok							
WB1641	Design Engineering Project 1	6					
WB1641 Toets 1	Projecttentamen	3					
WB1641 Toets 2	Groepswerk en individuele vaardigheden	3					
WB1642	Design Engineering Project 2	6					
WB1642 Toets 1	Projecttentamen	3					
WB1642 Toets 2	Groepswerk en individuele vaardigheden	3					
WB1643A	Design Engineering Project 3A	6					
WB1643A Toets 1	Projecttentamen	3					
WB1643A Toets 2	Groepswerk en individuele vaardigheden	3					
WB1643B	Design Engineering Project 3B	6					
WB1643B Toets 1	Projecttentamen	3					
WB1643B Toets 2	Groepswerk en individuele vaardigheden	3					
2e jaar BSc WB 2022							
WB201-11 Wiskunde blok	Mathematics block						
WBMT2048	Mathematics 3	6					
WBMT2048 Toets 1	Analyse - deeltentamen	3					
WBMT2048 Toets 2	Differentiaalvergelijkingen - deeltentamen	3					
WBMT2049	Mathematics 4	6					
WBMT2049 T1	Kansrekening en Statistiek - deeltentamen	3					
WBMT2049 T2	Numerieke Wiskunde - deeltentamen	3					
WBMT2049 T3	Numerieke Wiskunde - practicum	0					
Werktuigbouwkunde Theorie blok							
WB2235	Signal Analysis	6					
WB2330	Materials Sciences	6					
WB2542	Fluid Mechanics & Heat Transfer	6					
WB2542 Toets 1	Stromingsleer	3					
WB2542 Toets 2	Warmte-overdracht	3					
WB2630	Advanced Mechanics	6					
WB2630 Toets 1	Rigid-Body Dynamics	3					
WB2630 Toets 2	Continuum Mechanics	3					
Werktuigbouwkunde Project blok							
WB2232	Project Robotics	6					
WB2232 Toets 1	Robotics exam	3					
WB2232 Toets 2	Groepswerk	3					
WB2332	Project Materials Sciences	6					
WB2332 Toets 1	Materiaalkunde practicum + projecttoets	3					
WB2332 Toets 2	Groepswerk	3					
WB2543	Process Engineering & Thermodynamics	6					
WB2543 Toets 1	PE&T tentamen	3					
WB2543 Toets 2	Groepswerk + practicum	3					
WB2632	Advanced Engineering Design Project	6					
WB2632 Toets 1	AED tentamen + FEM opdracht	3					
WB2632 Toets 2	Groepswerk	3					
3e jaar BSc WB 2022							
3e jaar WB minors 2022	3rd Year WB minors 2022						
3e jaar WB major 2022	3rd Year WB major 2022						

WB3073	The engineer in society
WB3073 Toets 1	<i>I&S Exam</i>
WB3073 Toets 2	<i>Group Essay</i>
WB3135	Integrated Mechanical Systems
WB3135 Toets 1	<i>IMS - groepsopdracht</i>
WB3135 Toets 2	<i>IMS - tentamen</i>
WB3240	Systems and Control Engineering
WB3BEP-16	Bachelor End Project



Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Bachelor WB 2022	
Introduction 1	Please switch to Dutch language

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

1e jaar BSc WB 2022

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Wiskunde blok

WBMT1050	Mathematics 1	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	Dutch	

WBMT1050 Toets 1	Analyse 1 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	1	
Start Education	1	
Exam Period	1 2	

WBMT1050 Toets 2	Analyse 2 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	2	
Start Education	2	
Exam Period	2 3	

WBMT1051	Mathematics 2	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	Dutch	

WBMT1051 Toets 1	Lineaire Algebra 1 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3	
Start Education	3	
Exam Period	3 4	

WBMT1051 Toets 2	Lineaire Algebra 2 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	4 5	

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Werktuigbouwkunde Theorie blok

WB1135	Introductory Dynamics	6
Responsible Instructor	Prof.dr. P.G. Steeneken	
Instructor	R.F. Dooren	
Instructor	F.G.J. Broeren	
Instructor	Ir. A.M. van der Niet	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering 3mE Department Cognitive Robotics	

WB1530-14	Thermodynamics	6
Responsible Instructor	Dr. R. Pecnik	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB1630-20	Statics	6
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Course Coordinator	F.G.J. Broeren	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1631-15	Mechanics of Materials	6
Responsible Instructor	Dr.ir. M. Langelaar	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Werktuigbouwkunde Project blok

WB1641	Design Engineering Project 1	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	G. Radaelli	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1641 Toets 1	Projecttentamen	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	G. Radaelli	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Biomechanical Engineering	

WB1641 Toets 2	Groepswork en individuele vaardigheden	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	G. Radaelli	
Instructor	B. van Vliet	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	Dutch	
Department	3mE Department Biomechanical Engineering	

WB1642	Design Engineering Project 2	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1642 Toets 1	Projecttentamen	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1642 Toets 2	Groepswork en individuele vaardigheden	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643A	Design Engineering Project 3A	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643A Toets 1	Projecttentamen	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643A Toets 2	Groepswork en individuele vaardigheden	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643B	Design Engineering Project 3B	6
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643B Toets 1	Projecttentamen	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1643B Toets 2	Groepswork en individuele vaardigheden	3
Responsible Instructor	Dr.ir. R.W. Vroom	
Responsible Instructor	Dr.ir. A. van Beek	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

2e jaar BSc WB 2022

Prerequisites

To be able to participate in this course you must register in time.

Registration for a course is not the same as registration for a (partial) exam, registration in bright space or registration for a study link.

More information about registering for courses, the procedure and current deadlines can be found at;
<https://www.tudelft.nl/index.php?id=50357&L=1>.

Registration for exams must be done separately.

* in some cases there are entry requirements or additional criteria to be admitted to a course. These can be found in the TER,
<https://www.tudelft.nl/index.php?id=33610&L=1>.

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

WB201-11 Wiskunde blok

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics,	
Expected prior knowledge	First year Mathematics courses	
Course Contents	WBMT2048/T1: Series and sequences. Line and surface integrals. Integral Theorems of Green, Stokes and Gauss. WBMT2048/T2: Differential Equations (linear first and second order equations, Laplace transform, systems of linear equations, autonomous systems and stability, Fourier series, partial differential equations)	
Study Goals	Study goals for each part can be found on the corresponding CourseBase-page (WBMT2048/T1 or WBMT2048/T2).	
Education Method	Online Lectures, 4 hours per week.	
Literature and Study Materials	WBMT2048/T1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace. WBMT2048/T2: W.E. Boyce, R.C. DiPrima, D.B. Meade, Boyce's Elementary Differential Equations and Boundary Value Problems, global edition, 2017.	
Assessment	WBMT2048/T1: On-campus exams. Partly digital (in Grasple), partly written exams (see Brightspace for more details). There will be two partial exams: one partial exam in week 5 about the first part of the course (Sequences and Series) and one partial exam in week 10 about the second part of the course (Vector Calculus). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, about the complete course, that lasts 180 minutes. It is not possible to resit only one partial exam. If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace). WBMT2048/T2: On-campus exams. Partly digital (met Grasple), partly written (see Brightspace for more details). If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace).	
Exam Hours	WBMT2048/T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048/T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048/T1: A formula sheet and a flow chart series (see Brightspace). NO calculators. WBMT2048/T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	

WBMT2048 Toets 1	Analyse - deeltentamen	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics	
Expected prior knowledge	First year Mathematics courses	
Course Contents	Series and sequences. Line and surface integrals. Integral Theorems of Green, Stokes and Gauss.	
Study Goals	After this course the student should be able to 1. determine whether a sequence is convergent or divergent, for example by calculating its limit; 2. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 3. determine the interval of convergence of a power series; 4. determine the Taylor series of a simple analytic function, for example $\sin(x)$, $\exp(x)$, $\ln(x)$, $\arctan(x)$, $(1+x)^a$; 5. find parametrizations of curves and surfaces (in 2D and 3D); 6. calculate a line integral or a surface integral of a function or a vector field; 7. mention the physical interpretation of these line and surface integrals; 8. define and formulate the physical meaning of concepts like gradient, curl and divergence; 9. calculate line and surface integrals via one of the three integral theorems (Green, Stokes and Gauss).	
Education Method	Lectures, 4 hours per week (2x2).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital (in Grasple), partly written exams (see Brightspace for more details). There will be two partial exams: one partial exam in week 5 about the first part of the course (Sequences and Series) and one partial exam in week 10 about the second part of the course (Vector Calculus). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, about the complete course, that lasts 180 minutes. It is not possible to resit only one partial exam. If on-campus exams are not possible because of COVID-19 measures, the prescribed method of testing is: Online exams via Grasple (in Brightspace).	
Exam Hours	Each partial exam: 90 minutes. The resit: 180 minutes.	
Permitted Materials during Tests	A formula sheet and a flow chart series (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Remarks	This is half of the course WBMT2048. Partial grades for T1 and T2 will be given in one decimal. The final grade is a number rounded to the nearest half. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So the average is 6,4. The final grade will be: 6,5. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75).	
Judgement	See Dutch comment.	

WBMT2048 Toets 2	Differentiaalvergelijkingen - deeltentamen	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

WBMT2049	Mathematics 4	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3 4	
Start Education	3 4	
Exam Period	Different, to be announced	
Course Language	Dutch	

WBMT2049 T1	<i>Kansrekening en Statistiek - deeltentamen</i>	3
Responsible Instructor	Dr.ir. R. Versendaal	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Required for	Bachelor project, multivariate data analysis	
Expected prior knowledge	Techniques from first year math: integration, differentiation, the exponential function and the natural logarithm. Analyse 1, 2 and 3, Linear Algebra.	
Parts	The course is split in two parts: In the first part the basis of probability theory is presented with important distributions (such as uniform, normal, exponential, binomial, Bernoulli, geometric), random variables and their expectation and variance. Furthermore, joint distributions and their properties (such as covariance and correlation) are treated. In the second part it is explained how to summarize and interpret datasets, both graphically and numerically. Next, confidence intervals and hypothesis testing are treated.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebyshev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Lectures with interactive elements; pre-lecture videos, online exercises	
Computer Use	Python	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Prerequisites	Analysis 1, 2, 3, linear algebra	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one final exam (week 10) on Part 2: Statistics. Final grade is the average of both tests.	
Exam Hours	90 min	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Simple (non-graphical and non-programmable) calculator	

WBMT2049 T2	<i>Numerieke Wiskunde - deeltentamen</i>	3
Responsible Instructor	Dr. N.V. Budko	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

WBMT2049 T3	Numerieke Wiskunde - practicum	0
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See WBMT2049, Numerical mathematics part	
Study Goals	See WBMT2049, Numerical mathematics part	
Education Method	Practicum	
Practical Guide	One Jupyter Notebook per assignment must be completed and submitted. Notebooks can be submitted by a group of at most two students: these students must register as group in Brightspace (see above). Only students in a registered group are allowed to cooperate on the assignment. Notebooks can only be handed in through Assignments within Brightspace. Any notebook handed in any other way will not be assessed. Distributing to others and/or receiving from others the assignment and/or the resulting notebook is prohibited. With others we mean any party not being your group partner, the lecturers and/or the assistants. An action of one student counts as an action of the entire registered group. If any of the above rules are violated by a group, any assessment of the Notebooks of the violating group will be revoked.	
Assessment	On basis of a submitted Jupyter Notebook, resulting in either a V (pass) or an O (fail).	
Enrolment / Application	A separate electronic subscription for the numerical practicum might (also) required, see Brightspace for details. On top of that for participation in the course Mathematics 4 an early subscription for the module Mathematics 4 is required as well. More information regarding the general rules can be found at http://onderwijs.3me.tudelft.nl/osiris.	
Remarks	It concerns a partial score. Needed is a completion (V).	
Judgement	For rules concerning education you may consult http://onderwijs.3me.tudelft.nl/reglementen .	

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Werktuigbouwkunde Theorie blok

WB2235	Signal Analysis	6
Responsible Instructor	R. van de Plas	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	WI2030WBMT Wiskunde 3 (Differentiaalvergelijkingen) WI2031WBMT Wiskunde 4 (Kansrekening en Statistiek) should be heard in parallel to the course	
Course Contents	This course treats the analysis, filtering and detection of signals influenced by linear time-invariant systems as they occur in typical mechanical and mechatronic applications. Signals will be considered in both deterministic and stochastic formulations. Special emphasis is put on discrete-time techniques that can be easily implemented in practice. The main topics of the course are: 1. Elementary properties of signals and systems (even, odd, linear, stable, causal, linear-phase, group-delay, all-pass, minimum-phase) 2. Fourier and Hilbert transform, one and two-sided Laplace and Z-transforms 3. Bode plots and Bodes phase-gain relationship 4. Continuous-time low-pass filters 5. Modulation and demodulation 6. Analog-to-digital and digital-to-analog conversion 7. Design of discrete-time filters using windowing and impulse invariance 8. Elementary properties of random processes (i.i.d., w.s.s., joint w.s.s., independence); key indicators such as mean, auto and cross-correlation/covariance, and spectral densities 9. Identification of linear time-invariant discrete-time systems in the frequency domain 10. Discrete-time Wiener filtering (finite impulse response and non-causal) 11. Detection of discrete-time signals in Gaussian noise	
Study Goals	Students can 1. Evaluate if a signal or system possesses an elementary property; utilize these in simple problems; give examples that exhibit desired elementary properties 2. Compute Fourier, Laplace, z and Hilbert transforms, both directly and using transform pairs and properties 3. Sketch Bode plots for rational transfer functions 4. Determine the specifications of continuous-time low-pass filters from their frequency response; design low-pass filters that fulfill given specifications 5. Analyze typical modulation and demodulation schemes; add missing components 6. Evaluate the impact of A/D and D/A conversion schemes; choose sampling intervals 7. Design discrete-time filters using impulse invariance, windowing, and Wiener techniques 8. Evaluate if a random process possesses an elementary property; utilize them in simple problems; determine key indicators of random processes 9. Identify the frequency response of a linear time-invariant discrete-time system from actual measurements; specify the bias and the variance 10. Derive optimal detection rules for a known signal in Gaussian noise	
Education Method	Lectures, problem sets and practice sessions (werkcolleges).	
Literature and Study Materials	- A.V. Oppenheim, A.S. Willsky and S. Hamid, Signals and Systems, Pearson New International Edition, 2nd ed, 2013. - A.V. Oppenheim and G. Verghese, Signals, Systems and Inference, Class Notes for 6.011: Introduction to Communication, Control and Signal Processing, MIT, Spring 2010. Available online: http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-011-introduction-to-communication-control-and-signal-processing-spring-2010/readings/ - For some parts of the lecture, external materials will be used. These materials will either be available online from within the network of TU Delft, or be provided via Brightspace.	
Assessment	Written exam. Students will only be allowed to bring a non-programmable and non-graphing calculator. A formula sheet will be provided.	
Department	3mE Department Delft Center for Systems and Control	

WB2330	Materials Sciences	6
Responsible Instructor	Dr.ir. S.E. Offerman	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Materials Science & Engineering	

WB2542	Fluid Mechanics & Heat Transfer	6
Responsible Instructor	Prof.dr.ir. C. Poelma	
Responsible Instructor	Dr.ir. M.J. Tummers	
Instructor	Dr.ir. G.E. Elsinga	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB2542 Toets 1	<i>Stromingsleer</i>	3
Responsible Instructor	Prof.dr.ir. C. Poelma	
Instructor	Dr.ir. G.E. Elsinga	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB2542 Toets 2	<i>Warmte-overdracht</i>	3
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB2630	<i>Advanced Mechanics</i>	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Responsible Instructor	Prof.dr. H. Vallery	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	3mE Department Biomechanical Engineering 3mE Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2630 Toets 1	Rigid-Body Dynamics	3
Responsible Instructor	Prof.dr. H. Vallery	
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hourse 2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Expected prior knowledge	Prerequisites for the course: see document on Brightspace or navigate to dbl.tudelft.nl -> Student Resources->preparatory material for the course "Rigid-Body Dynamics" Note that the first test will also cover prerequisites.	
Course Contents	1) Kinematics of a rigid body in 3D a. Description of single and successive 3D rotations via Euler angles, rotation matrices, axis and angle of rotation. b. Angular velocity and instantaneous axis of rotation c. Transport Theorem 2) Kinetics of a rigid body in 3D a. The inertia tensor b. Principle of work & energy c. Newton-Euler equations d. Free rotational motion, gyroscopic effects 3) Systems with n degrees of freedom a. Generalized coordinates b. Principle of virtual work c. Lagranges equations 4) Methods to check plausibility of generated results, such as a. Units notation and checks b. Numerical approximation c. Checking validity of assumptions d. Treatment of special cases	
Study Goals	Kinematics of a rigid body in 3D The student is able to describe rotations of a rigid body in 3D via Euler angles, rotation matrices, and the principal axis of rotation. transform one description of rotations into the other, and re-write successive rotations as a single rotation in the different representations. describe angular velocity of a rigid body in 3D and find the instantaneous axis of rotation Kinetics of a rigid body in 3D The student is able to calculate the inertia tensor for a rigid body via integration analyze the inertia tensor and find the principal axes derive the equations of motion for a rigid body in 3D using the principle of work and energy or using the Newton-Euler equations recognize and explain gyroscopic effects Systems with n degrees of freedom The student is able to describe general motion of systems of particles and of rigid bodies in generalized coordinates in 2D and 3D apply the principle of virtual work to derive the equations of motion for systems of particles and rigid bodies in 2D and 3D formulate kinetic and potential energy for systems of particles and for rigid bodies in generalized coordinates apply Lagranges methods to systems of particles and to rigid bodies to obtain the equations of motion Recognize conservative systems Generating reliable results to mechanical problems The student is able to test the plausibility of calculation steps and final results be aware of and check validity of assumptions whenever applying learned concepts	
Education Method	Students prepare the course material themselves each week. For this, they can use the book and material provided on Brightspace. On Brightspace, also an indication of the weekly topics will be provided. Each week, interactive on-campus lectures and on-campus tutored sessions are organized. During the tutored sessions, students can work on homework exercises and ask for further explanation and help. The first week, a formative digital test is provided to help students understand their current level with respect to prerequisites. In contrast to the mid-term and the final exam (see below), this test does not count towards the courses grade.	
Books	Prerequisites for the course: see document on Brightspace or navigate to dbl.tudelft.nl-> Student Resources->preparatory material for the course "Rigid-Body Dynamics" Vallery/Schwab "Advanced Dynamics", in 1st (black), 2nd (blue), or 3rd (red) edition	
Assessment	additional material: Greenwood Advanced Dynamics, Cambridge University Press, 2006 The main assessment of this course is a final on-campus exam, taken individually. The exam is taken digitally, in computer rooms. A small bonus can be obtained via an additional individual mid-term exam, which has the same form as the final exam. The final exam and the mid-term are organized on-campus. They are proctored and open-book, which means that students may bring their book (Vallery&Schwab and/or Greenwood, see above.) and 1 double-sided A4 sheet of hand-written notes. The A4 sheet must be written by the student him/herself and have the students name printed on it. The final grade G for the course is calculated from the midterm grade M and the exam grade E as follows: If $E < 5.5$, then the final grade $G=E$. If $E \geq 5.5$, then the final grade $G=0.9 \cdot E + 0.1 \cdot \max(M,E)$. Three examples to clarify the calculation of the grades: 1. Student 1 obtained the grades: Mid-term $M=7$, final exam grade $E=5$. As $E < 5.5$, the final grade $G=E=5$. 2. Student 2 did not participate in the mid-term, but obtained a final exam grade of $E=8$. The students final grade is $G=E=8$.	

	3. Student 3 participated in the mid-term and obtained $M=8$, and a final exam grade $E=5.5$. The students final grade is $G=0.9 \cdot E + 0.1 \cdot M = 5.8$. The mid-term exam is only offered in Q1, and its grade is only valid for this academic year. The resit of the course contains only a digital final exam.
Enrolment / Application	See Dutch version
Department	3mE Department Biomechanical Engineering
Judgement	See Dutch version
Contact	Rigid-Body-Dynamics-3me@tudelft.nl. Please do not use lecturers' individual e-mail addresses.

WB2630 Toets 2	Continuum Mechanics	3
Responsible Instructor	Dr.ir. L.F.P. Noel	
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hourse 2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	1) Linear 3D continuum theory - Deformation tensor in 3D, linear - Stretch in arbitrary direction - Stress vector and stress tensor in 3D, linear - Equilibrium in 3D - Principal stress directions, shear stress acting on a plane - Generalized Hooke's law: isotropic, anisotropic, homogeneous, heterogeneous 2) Energy principles - PVW without deformations - PVW with deformation - Minimum of potential energy 3) Systems with n-dof - Introduction to a systematic way of solving n-dof problems. The discussion will be based on simple spring or truss problems. Aspects: DOFs, generalized deformations, generalized stresses, generalized constitutive equations, discrete continuity and equilibrium, system assembly, system matrix, boundary conditions, solution 4) Finite Element Method - Approximations based on minimum of potential energy: - Shape or interpolation functions - Natural coordinate systems - Displacement based finite elements - Derivation of simple finite element - Mass matrix + eigenfrequencies - Topics required to use FEM: - Distributed loads - Local coordinates - Convergence, mesh density and mesh refinement - Kinematic coupling - Element choice - Different disciplines - Frequently made mistakes 5) Introduction to plates - Introduction to plates (in-plane as well as out-of-plane) - Kinematic model for plates, deformations of the reference plane - Linear constitutive equations for laminated plates (ABD matrix) - Link with minimum of potential energy - FEM NOT included are dynamic BC, e.g. Kirchhoff equivalent loads and loads at corners	
Study Goals	Linear 3D continuum theory The student is able to: - Recognize problems for which continuum theory is not applicable - Explain and use the governing equations in 3D: continuity, equilibrium/eqs. of motion - Determine for known displacement fields the deformations, and can identify the limitations - Calculate deformations and stretch in a specified direction - Use and explain a stress description in 3D and is able to determine a stress vector - Determine shear stress, normal stress and principle stress and directions, describe the practical relevance - Apply generalized Hooke's law, describe its relevance and limitations Energy principles The student is able to: - Describe PVW and apply it to simple problems to derive equilibrium equations - Describe MPE and apply it to simple problems to build approximate solutions - Recognize when PVE and MPE can be applied correctly Systems with n-dof: The student is able to: - Describe the solution approach for simple and linear n-DOF problems from a global perspective. - Set up a set of equations manually for relatively simple and linear problems - Solve problems with few DOF entirely FEM: The student is able to: - Describe the basics of the finite element method - Describe the basic steps required for the formulation of a simple finite element - Perform a linear FEM analysis with a commercial FEM package (although training for a specific FEM package will be required) Plates: The student is able to: - Give a global description of plate/shell models - Recognize when plate/shell models are preferred. - Specify kinematic boundary conditions - Use min of potential energy to obtain simple approximations for deflections, stresses and deformations for laminated plates	
Education Method	Contact hours: - Integrated plenary sessions, including lecture, exercises, and in-class tests (4 hours per week) - Weekly homework - Question hours (2 hours per week)	

In-class tests:
 In-class tests provide bonus credits
 Bonus credits are only valid for one academic year. Thus, the bonus credits evaporate at the end of the academic year
 During the in-class tests you may collaborate with your neighbor students and use internet info, slides, reader, books.
 Use of any type of electronic communication (sms, cell phone, email, facebook, whatsapp, etc etc.) is considered exam fraud and will be treated as such

Grade calculation:
 Provided the grade of the written exam is 5.00 or higher: $\text{Final-Grade} = 0.5 * (\text{Grade-written-exam}) + 0.5 * \text{MAX}(\text{Grade-written-exam}, \text{Grade-inclass-tests})$
 Grade-inclass-tests is based on average of best 70% of individual test grades. No participation in a test leads to a 0 as grade for that particular test.
 In case the grade of the written exam is lower than 5.00, one can not benefit from the bonus credits. Example: the grade of the written exam is 4.95, in that case there can be no benefit from the bonus credits

Study materials:
 Reader (available in pdf format)
 Mechanics of Materials, Hibbeler, (Buckling Chp. 13, Energy Methods Chp. 14)
 Concepts and applications of finite element analysis, Robert Cook et al., Wiley could be used for further reading

Assessment

Write exam (bonus credits can be obtained with in-class tests)

Special Information

In the academic year 2017-2018 the lectures have been presented in English.

Department

3mE Department Precision & Microsystems Engineering

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

Werktuigbouwkunde Project blok

WB2232	Project Robotics	6
Responsible Instructor	M. Wisse	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	Please consult the course information in Dutch.	
Study Goals	Please consult the course information in Dutch.	
Education Method	Please consult the course information in Dutch.	
Books	"Electronics - a systems approach" (6th edition), Neil Storey	
Assessment	Please consult the course information in Dutch.	
Enrolment / Application	See Dutch version for rules of enrolment.	
Remarks	The course runs tightly integrated with the Signal Analysis course, thus students are expected to follow both courses simultaneously.	
Department	3mE Department Cognitive Robotics	
Judgement	See http://onderwijs.3me.tudelft.nl/reglementen	

WB2232 Toets 1	Robotics exam	3
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	See general description for detailed info.	
Study Goals	See general description for detailed info.	
Education Method	Lectures	
Assessment	Written (computer) exam and weekly lab practicals.	
Enrolment / Application	See Dutch version	
Remarks	This is a part of course. The grade for this part is computed as 60% from the computer exam, whenever more than 45% of the exam points are achieved, and 40% from the weekly practicals (4 best out of 5). Partial grades are given with one decimal. The minimum to average with the other part of the course is a 5,0. Example of averaging: 5,0 for part 1 grade + 6,5 for part 2 grade = $11,5:2 = 5,75 = 6,0$ 5,75 is the minimum needed for a valid final grade.	
Department	3mE Department Cognitive Robotics	
Judgement	See Dutch version	

WB2232 Toets 2	Groepswork	3
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See course description for details.	
Study Goals	See course description for details.	
Education Method	Project	
Assessment	Project development in groups of 4. Evaluated based on a presentation/demo (50%) and a report (50%)	
Enrolment / Application	See Dutch version.	
Remarks	This is a part of a course.	
Department	3mE Department Cognitive Robotics	
Judgement	See Dutch version	

WB2332	Project Materials Sciences	6
Responsible Instructor	Prof.dr. J. Dik	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch English	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Education Method	See Dutch description	
Assessment	See Dutch description	
Enrolment / Application	See Dutch description	
Department	3mE Department Materials Science & Engineering	
Judgement	See Dutch description	

WB2332 Toets 1	Materiaalkunde practicum + projecttoets	3
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Department	3mE Department Materials Science & Engineering	

WB2332 Toets 2	Groepswork	3
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	See Dutch description	
Study Goals	See Dutch description	
Department	3mE Department Materials Science & Engineering	

WB2543	Process Engineering & Thermodynamics	6
Responsible Instructor	Dr.ir. J.T. Padding	
Instructor	Dr. R. Delfos	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Tags	Design Energy Energy & Industry Fluid Mechanics Group work Modelling Practicals Process Project Sustainability Technology Thermodynamics Transport phenomena	
Department	3mE Department Process & Energy	

WB2543 Toets 1	<i>PE&T tentamen</i>	3
Responsible Instructor	Dr.ir. J.T. Padding	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB2543 Toets 2	<i>Groepswerk + practicum</i>	3
Responsible Instructor	Dr. R. Delfos	
Responsible Instructor	Dr.ir. J.T. Padding	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB2632	<i>Advanced Engineering Design Project</i>	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	8.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Remarks	Hoorcollege online 4/0/0/0 Projecturen 4/0/0/0	
Department	3mE Department Precision & Microsystems Engineering	

WB2632 Toets 1	<i>AED tentamen + FEM opdracht</i>	3
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB2632 Toets 2	<i>Groepswerk</i>	3
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

3e jaar BSc WB 2022

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

3e jaar WB minors 2022

Year	2022/2023
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Mechanical Engineering

3e jaar WB major 2022

WB3073	The engineer in society	4
Responsible Instructor	Dr. E. Ziliotti	
Contact Hours / Week x/x/x/x	0/x/0/x	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4 5	
Course Language	English	
Course Contents	In this course you will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to a range of relevant aspects and concepts, including the distinction between descriptive and normative reasoning, philosophical and professional ethics, ethics of robotics, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse ethical, legal and social problems of technology in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems. You will write a text on ethical, social or legal aspect of technology.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. can embed ethical and social reasoning into a technical project are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way. can write a short academic essay of critical analysis of an ethical or societal issue related to technology	
Education Method	A series of 7 lectures and the writing of an essay on ethical, social or legal aspects of technology to be written under the supervision of instructors from the section Ethics of Technology at TBM. The essay will be related to and included into the bachelor end project.	
Literature and Study Materials	Articles and documents posted on Brightspace; lectures (slides)	
Assessment	A digital exam (50%) and a written essay (50%) on the ethical, social or legal aspect of technology to be written under the supervision of a lecturer from the Section Ethics/Philosophy to Technology at TBM; the essay will be part of the bachelor end project. Exam and essay can be done in Q2 or Q4. In order to pass the course both the digital exam and the essay should be graded with at least 5 (and the average should be sufficient).	
Enrolment / Application	To be able to participate in this course you must register in time. Registration for a course is not the same as registration for a (partial) exam, registration in bright space or registration for a study link. More information about registering for courses, the procedure and current deadlines can be found at; https://www.tudelft.nl/index.php?id=50357&L=1 . Registration for exams must be done separately. * in some cases there are entry requirements or additional criteria to be admitted to a course. These can be found in the TER, https://www.tudelft.nl/index.php?id=33610&L=1 .	

WB3073 Toets 1	I&S Exam	2
-----------------------	---------------------	----------

WB3073 Toets 2	Group Essay	2
-----------------------	--------------------	----------

WB3135	Integrated Mechanical Systems	6
Responsible Instructor	Dr.ir. P. Breedveld	
Instructor	Dr.ir. J.F.L. Goosen	
Instructor	Dr.ir. D.L. Schott	
Instructor	B. van Vliet	
Instructor	Prof.dr.ir. B.J. Boersma	
Instructor	Prof.dr.ir. S.A. Klein	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Biomechanical Engineering	

WB3135 Toets 1	IMS - groepsopdracht	3
Responsible Instructor	Dr.ir. P. Breedveld	
Instructor	Dr.ir. D.L. Schott	
Instructor	Dr. R. Pecnik	
Instructor	F.J. Goosens	

WB3135 Toets 2	IMS - tentamen	3
Responsible Instructor	Dr.ir. P. Breedveld	

WB3240	Systems and Control Engineering	6
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Instructor	S.C. Bregman	
Instructor	Dr.ir. S.P. Mulders	
Instructor	Dr.ir. K. Batselier	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Delft Center for Systems and Control	

WB3BEP-16	Bachelor End Project	14
Responsible Instructor	Dr. R. Delfos	
Course Coordinator	Dr. S. van de Velde	
Instructor	Ir. M.B. Duinkerken	
Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Dr. D. Dodou	
Instructor	J.W. Haverkort	
Instructor	Dr.ir. M.B. Mendel	
Instructor	Dr. G.J. Verbiest	
Instructor	Dr. P. Dey	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 3	
Course Language	Dutch	
Education Method	See: http://www.wbmt2.tudelft.nl/bep-wb/BEP_def_EN.pdf	
Department	3mE Department Biomechanical Engineering 3mE Department Delft Center for Systems and Control 3mE Department Maritime & Transport Technology 3mE Department Materials Science & Engineering 3mE Department Precision & Microsystems Engineering 3mE Department Process & Energy 3mE Department Cognitive Robotics	

Dr.ir. K. Batselier

Unit	Mech, Maritime & Materials Eng
Department	Team Kim Batselier
Telephone	+31 15 27 82725

Dr.ir. A. van Beek

Unit	Mech, Maritime & Materials Eng
Department	Mechatronic Systems Design
Telephone	+31 15 27 86984

Prof.dr.ir. B.J. Boersma

Unit	Mech, Maritime & Materials Eng
Department	Marine and Transport Technology
Telephone	+31 15 27 87979
Room	34.B-1-320

Dr.ir. P. Breedveld

Unit	Mech, Maritime & Materials Eng
Department	Medical Instruments & Bio-Inspired Technology
Telephone	+31 15 27 85232
Room	34.E-3-340

S.C. Bregman

Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 84993
Room	34.C-1-300

F.G.J. Broeren

Department	Precision and Microsystems Engineering
Telephone	+31 15 27 85905
Room	34.G-1-465

Dr. N.V. Budko

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 86050
Room	36.HB 03.050

Dr. R. Delfos

Unit	Mech, Maritime & Materials Eng
Department	Energy Technology
Telephone	+31 15 27 82963
Room	34b.K-0-270

Dr. P. Dey

Unit	Mech, Maritime & Materials Eng
Department	Team Poulumi Dey

Prof.dr. J. Dik

Unit	Mech, Maritime & Materials Eng
Department	Materials Science and Engineering
Telephone	+31 15 27 89571

Dr. D. Dodou

Unit	Mech, Maritime & Materials Eng
Department	Medical Instruments & Bio-Inspired Technology
Telephone	+31 15 27 84221
Room	34.E-1-270

R.F. Dooren

Department	Precision and Microsystems Engineering
Telephone	+31 15 27 84659

Department	Precision and Microsystems Engineering
Telephone	+31 15 27 84659

Ir. M.B. Duinkerken

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 81790
Room	34.B-3-320

Dr.ir. G.E. Elsinga

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 88179
Room	34b.K-2-270

Dr.ir. J.F.L. Goosen

Unit	Mech, Maritime & Materials Eng
Department	Structural Optimization and Mechanics
Telephone	+31 15 27 86500

F.J. Goosens

J.W. Haverkort

Unit	Mech, Maritime & Materials Eng
Department	Energy Technology
Telephone	+31 15 27 86651
Room	34b.K-0-260

Prof.dr.ir. J. Hellendoorn

Department	College van Bestuur
Telephone	+31 15 27 89007
Room	34.F-2-440

Unit	Mech, Maritime & Materials Eng
Department	Cognitive Robotics
Telephone	+31 15 27 89007
Room	34.F-2-440

Prof.dr.ir. J.L. Herder

Unit	Mech, Maritime & Materials Eng
Department	Precision and Microsystems Engineering
Telephone	+31 15 27 84713
Room	34.G-1-270

Prof.dr.ir. S.A. Klein

Department	Energy Technology
Telephone	+31 15 27 85840

Dr.ir. M. Langelaar

Unit	Mech, Maritime & Materials Eng
Department	Structural Optimization and Mechanics
Telephone	+31 15 27 86506
Room	34.G-1-300

S.M. van der Meer

Unit	Mech, Maritime & Materials Eng
Department	MSE Support
Telephone	+31 15 27 83976
Room	34.H-2-320

Dr.ir. M.B. Mendel

Unit	Mech, Maritime & Materials Eng
Department	Team Bart De Schutter
Telephone	+31 15 27 87539

Dr.ir. S.P. Mulders

Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 89189
Room	34.C-2-270

Dr. N.J. Myers

Department	Team Nitin Myers
Telephone	+31 15 27 87130

Ir. A.M. van der Niet

Department	Support Cognitive Robotics
Room	34.F-2-420

Dr.ir. L.F.P. Noel

Department	Computational Design and Mechanics
Telephone	+31 15 27 87467

Dr.ir. S.E. Offerman

Unit	Mech, Maritime & Materials Eng
Department	Team Erik Offerman
Telephone	+31 15 27 82198

Dr.ir. R.A.J. van Ostayen

Unit	Mech, Maritime & Materials Eng
Department	Mechatronic Systems Design
Telephone	+31 15 27 81647
Room	34.G-1-460

Ir. D. den Ouden-van der Horst

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Room	36.HB 03.250

Dr.ir. J.T. Padding

Unit	Mech, Maritime & Materials Eng
Department	Complexe Vloeistof Processen
Telephone	+31 15 27 86605
Room	34b.K-1-200

Dr. R. Pecnik

Department	Energy Technology
Telephone	+31 15 27 89153
Room	34b.K-1-220

R. van de Plas

Unit	Mech, Maritime & Materials Eng
Department	Team Raf Van de Plas
Telephone	+31 15 27 84106
Room	34.C-1-260

Prof.dr.ir. C. Poelma

Unit	Mech, Maritime & Materials Eng
Department	Process and Energy
Telephone	+31 15 27 82620
Room	34b.K-0-120

G. Radaelli

Unit	Mech, Maritime & Materials Eng
Department	Mechatronic Systems Design
Telephone	+31 15 27 86065
Room	34.G-1-160

Dr.ir. D.L. Schott

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 83130
Room	34.B-4-300

Prof.dr. P.G. Steeneken

Department	QN/Steeneken Lab
Telephone	+31 15 27 89545
Room	34.G-1-390

Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 89545
Room	34.G-1-390

Unit	Technische Natuurwetenschappen
Department	QN/Steeneken Lab
Telephone	+31 15 27 89545
Room	34.G-1-390

Dr.ir. R.F. Swarttouw

Department	Analysis
Telephone	+31 15 27 83634
Room	36.HB 08.290

Dr.ir. M.J. Tummers

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 82477
Room	34b.K-2-270

Prof.dr. H. Vallery

Unit	Mech, Maritime & Materials Eng
Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 83517
Room	34.E-1-320

Dr. S. van de Velde

Department	Onderwijs en Studentenzaken
Telephone	+31 15 27 82350

Dr. G.J. Verbiest

Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 86504

Dr.ir. R. Versendaal

Department	Applied Probability
Telephone	+31 15 27 87261
Room	36.HB 07.290

Department	Applied Probability
Telephone	+31 15 27 87261
Room	36.HB 07.290

B. van Vliet

Unit	Mech, Maritime & Materials Eng
Department	Support Biomechanical Engineering
Telephone	+31 15 27 83436
Room	34.E-3-330

Prof.dr.ir. T.J.H. Vlugt

Unit	Mech, Maritime & Materials Eng
Department	Engineering Thermodynamics
Telephone	+31 15 27 87551
Room	34b.K-0-150

Dr.ir. R.W. Vroom

Unit	Mech, Maritime & Materials Eng
Department	Mechatronic Systems Design
Telephone	+31 15 27 81342
Room	34.G-1-160

Prof.dr.ir. J.W. van Wingerden

Unit	Mech, Maritime & Materials Eng
Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 81720
Room	34.C-2-340

M. Wisse

Unit	Mech, Maritime & Materials Eng
Department	Robot Dynamics
Telephone	+31 15 27 86834
Room	34.F-2-080

Dr. E. Ziliotti

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 82139
